Case Study Report 1 – Exercise 3

Attribution

The dataset used in this report is sourced from ABS statistics for Retail Trade, Australia (Australian Bureau of Statistics, 2024). The focus will be particularly on the Original-adjusted data for the Total State Department Stores Turnover (Series ID: A3348618X as provided in Table 1: Retail Turnover, By Industry Group.

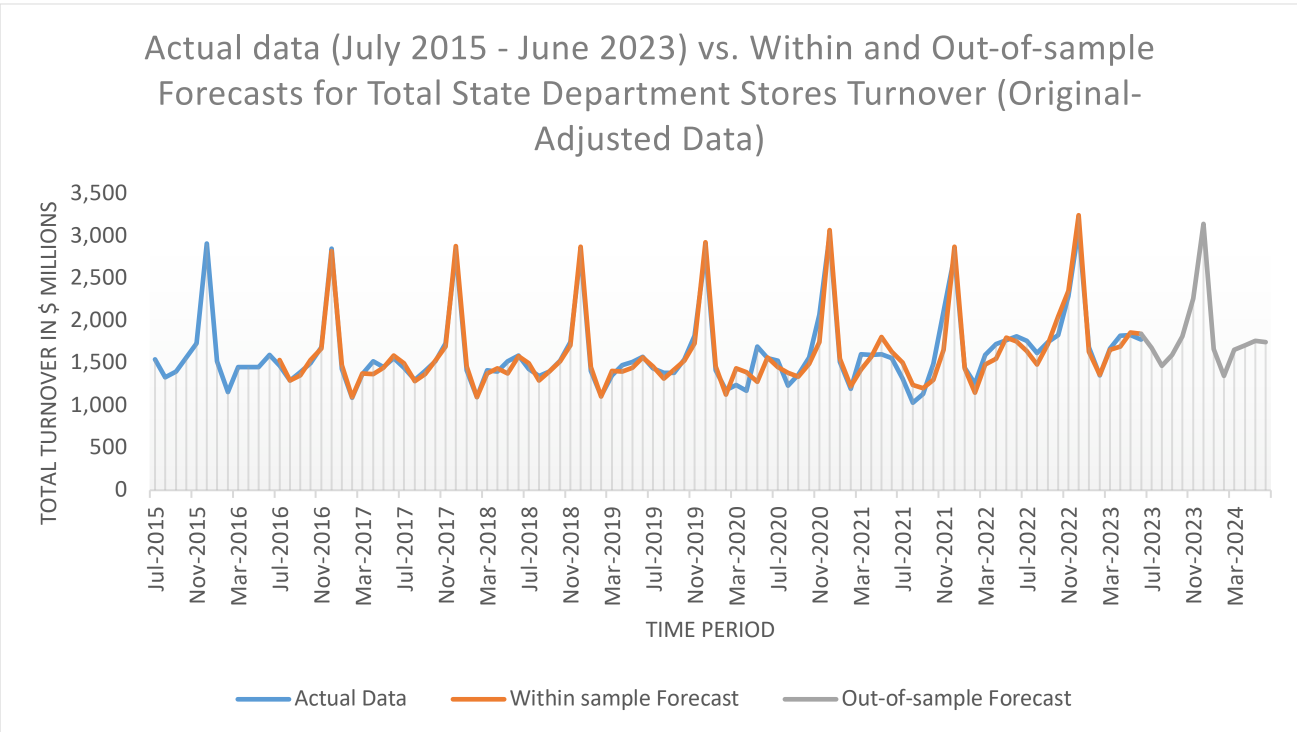
Scope

Sales of Australian department stores fluctuate due to factors like holidays, major sales and events, with December consistently showing the highest turnover. This report analyses seven years of monthly turnover data (July 2015 - June 2023) to forecast turnover trends for the upcoming 12-month period, from July 2023 to June 2024. Our goal is to generate predictions that reflect the real-world patterns to support future business planning.

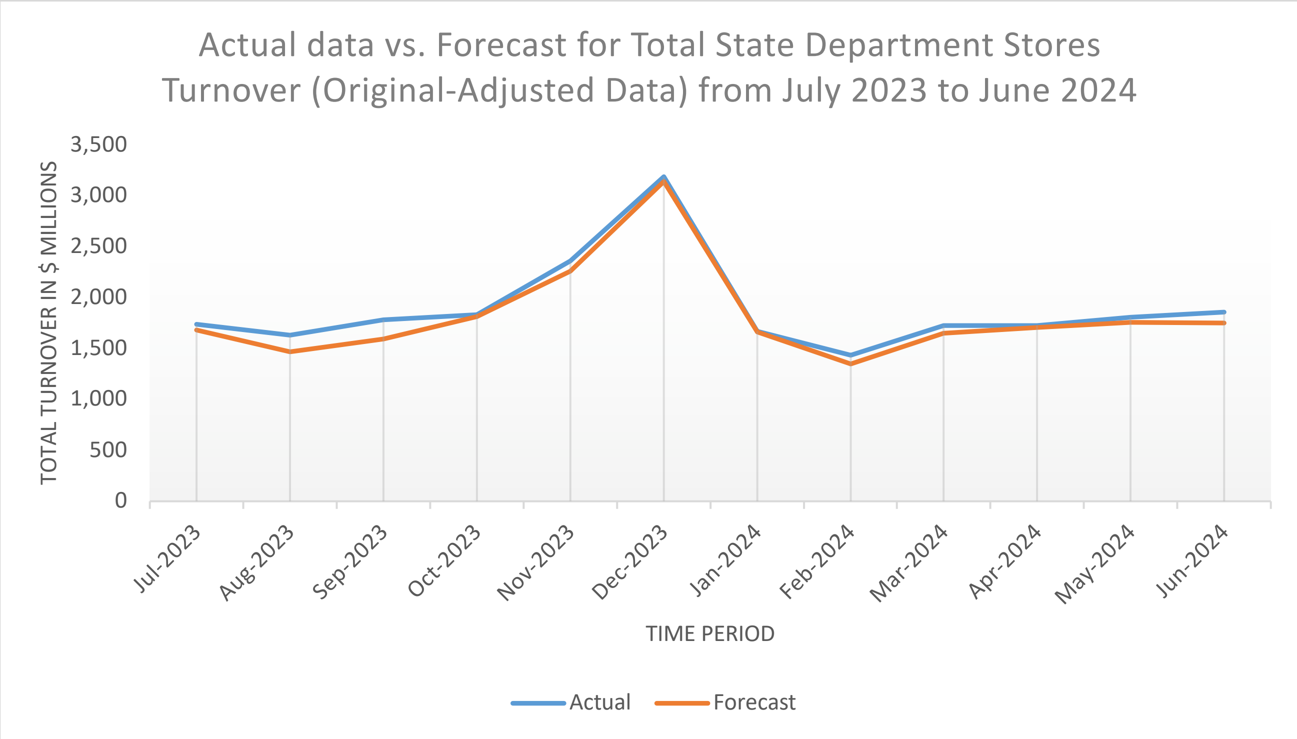
Application

The graph shows a systemic, repetitive fluctuation, with spikes in December, indicating a seasonal component. These fluctuations are constant with peaks and troughs confined within a parallel band, demonstrating additive seasonality. The Winter Exponential Smoothing (WES) model is the best fit for forecasting this type of component. This model uses seasonality equations and 3 smoothing parameters for level (alpha), trend (beta), and seasonality (gamma). To enhance accuracy, we will aim to minimize error metrics like Mean Squared Error (MSE) by changing the parameter values. The Excel function "Solver" can generate the optimum combination of the smoothing parameters to get the best accurate forecast.

Analysis

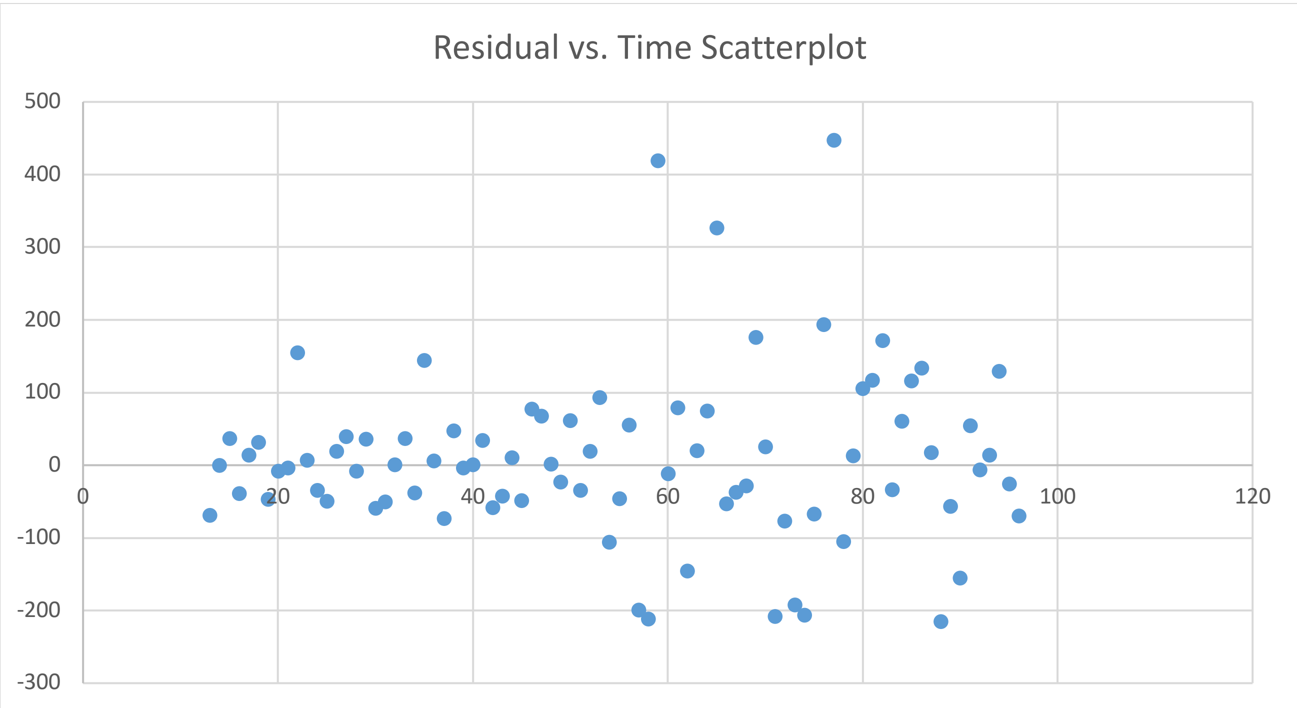


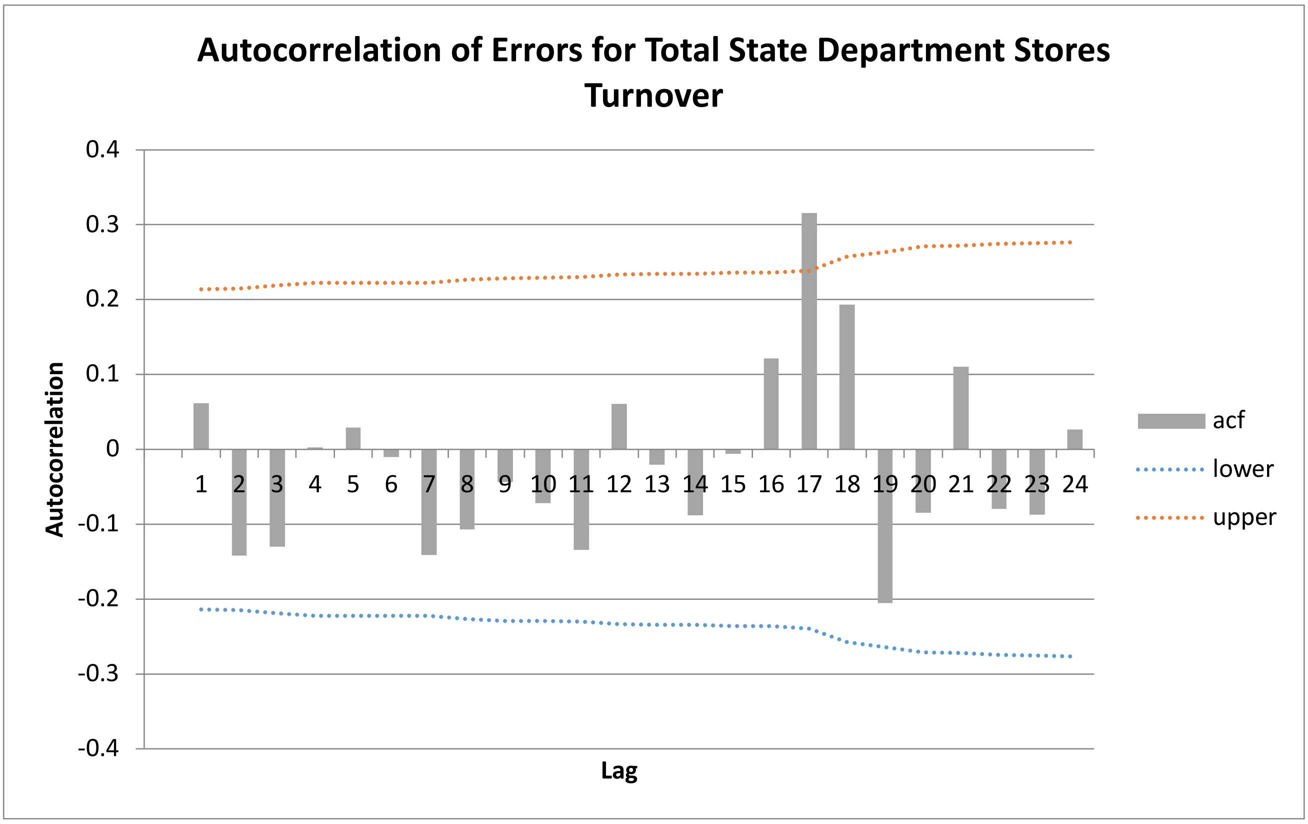
The line graph compares the original data to the forecast using the WES model. Here, the post-optimized data is used, where the smoothing parameters are already adjusted to generate the smallest MSE possible. The blue line depicts the actual data from July 2015 to June 2023. The WES model requires data from the first year to make forecasts, hence the within-sample forecast (orange line) starts a year after the starting period of actual data, from July 2016 to June 2023. The out-of-sample forecast (grey line) covers July 2023 to March 2024. The graph reveals regular and drastic peaks around $3,000 million in December and troughs around $1,100 million in February every year. The close alignment between the actual and within-sample forecast suggests high accuracy, though some variation can be seen around 2020 and 2021. The out-of-sample forecast shows a similar seasonal pattern with the past period forecast.



This line graph exhibits the comparison between actual data and forecast for July 2023 to June 2024. The patterns are quite similar, with minor deviations. Notably, forecasts for January and April 2024 show minimal error, less than $20 million. The peak in December closely matches the actual data. However, the bigger errors were found in August and September 2023, exceeding $150 million.

<u>Articulation of Issues</u>





To evaluate the model validity and error randomness, two methods are used. First, the scatter plot of residual versus time shows no clear pattern, implying randomness. There are a few outliers around residuals ranging from 300 to 500, but they do not have much influence on overall randomness. Second, the correlogram indicates that most autocorrelation function (ACF) values lie within the confidence intervals, supporting that residuals are generally random. However, autocorrelation at Lag 17 (0.316) surpasses the upper confidence interval. Overall, most residuals appear to be random, suggesting that the WES model is appropriate for this dataset.

Critique

The WES model generates the forecasts closely aligned with actual data, effectively capturing additive seasonal patterns such as the December peak and February trough. However, a thorough examination finds some significant errors, including a few outliers in scatter plot and one large autocorrelation, indicating that the errors are mostly random, but not perfectly random. Decomposition models may be able to manage seasonal patterns and respond to changes more effectively than WES by eliminating seasonality and randomness, but it can be complex to implement and interpret.

Regardless of how accurate the model is, forecasts cannot guarantee full accuracy since they are based on past data. External factors such as economic conditions (e.g., inflation and recession) and unforeseen global events (e.g., pandemics and political situations) can have a considerable impact on consumer spending behaviour, product availability and sales. While the models should ideally account for potential disruptions, incorporating such factors can be challenging, and at times, impossible.

Position

The WES model forecast provides useful insights into seasonal trends, highlighting peaks and troughs which can be used to guide strategic business decisions. Understanding these patterns helps businesses to plan stocks, marketing initiatives, and sales promotions more efficiently. However, while the WES model provides a solid foundation, minor errors suggest the model may not fully represent all market dynamics. To improve prediction accuracy, businesses should consider risk and scenario planning to anticipate potential disruptions. Furthermore, refreshing the dataset with new data, regular updates and monitoring on the model can improve adaptability and relevance. Combining these tactics allows businesses to reflect their forecasts better with actual market conditions and make better decisions.